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3/20/25

LAB 7

Optocouplers and Relay H Bridge

Introduction

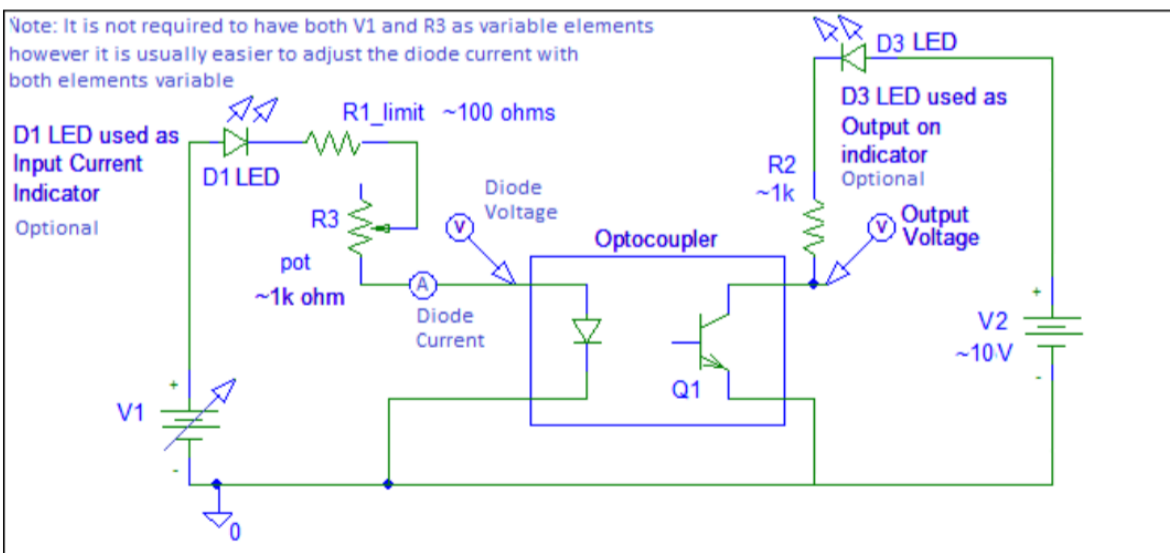
In this lab, we explored the function and applications of optocouplers—devices that provide electrical isolation between input and output signals using light as the transmission medium. Optocouplers are widely used in industrial systems, particularly as input interfaces to Programmable Logic Controllers (PLCs), where they serve to protect sensitive control circuitry from high voltages or electrical noise.

We examined three types of optocouplers: integrated circuit (IC) style optoisolators, slotted (opto-interrupter) devices, and reflective optocouplers. Each of these devices uses a light-emitting diode (LED) and a photodetector in different configurations to transmit signals while maintaining electrical separation. In the lab, we constructed basic test circuits for each optocoupler type, using a power supply and current-limiting resistor to drive the LED emitter, and analyzed the open-collector output behavior on the phototransistor side. This hands-on experience helped us understand the practical operation and advantages of optocouplers in real-world applications..

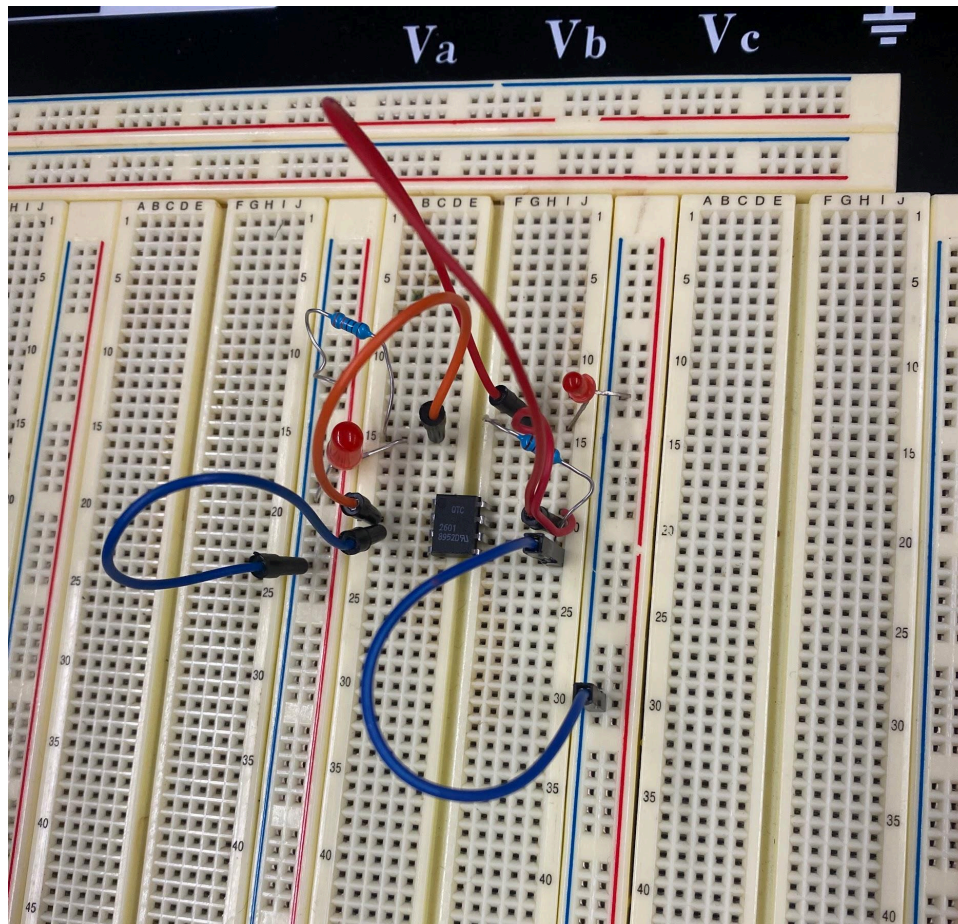
Optocouplers used in this lab:

- A) Optoisolators, IC style: 4N36, ICPL1631, H11L1
- B) Slotted type or Opto-interrupters: H22A1, OPB970, OPB854A1 or OPB811L55,
- C) Reflective, no number but similar to the Honeywell HOA0149 unit, custom wired.

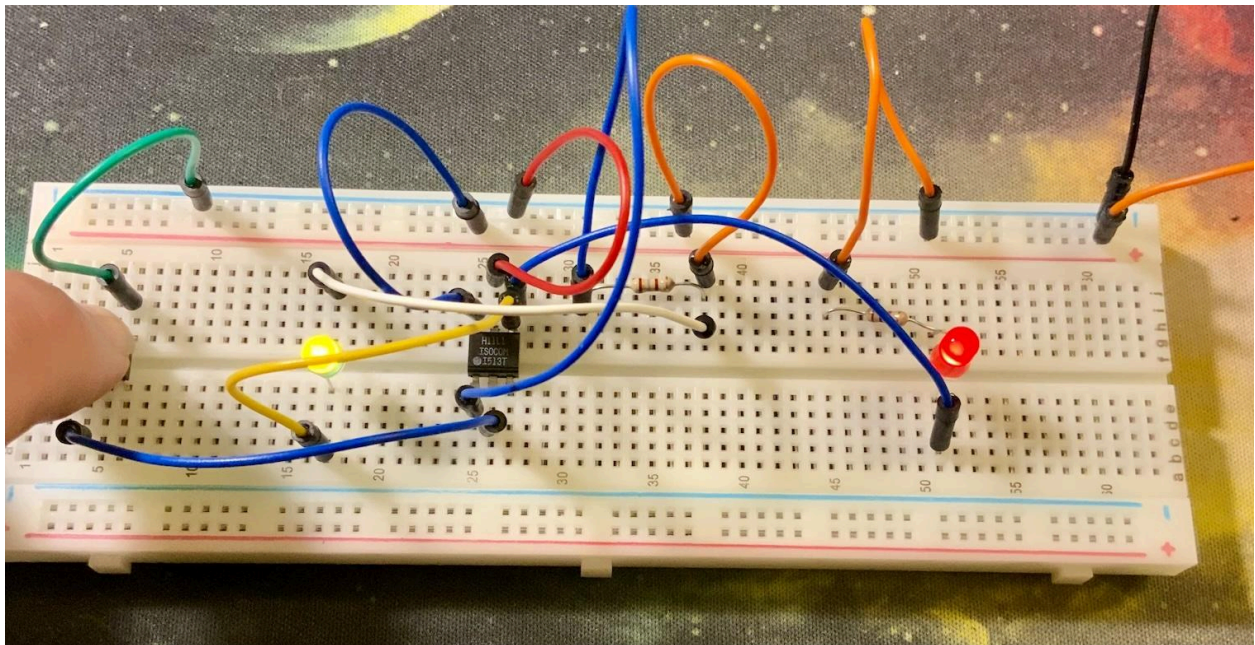
The test circuit for each of the optocouplers is shown below and remains similar for each type of optocoupler. It is important to note that this is not a typical application circuit– the input circuit of the optocoupler is typically only a resistor (or sometimes a switch with a resistor), but the DI LED indicates visually that there is a diode current. The potentiometer in the circuit (R3) allows for an adjustable input current. Note that using a potentiometer without some sort of safety net to protect the potentiometer is unwise. This is why we place resistor R1 before R3, to be used as a current limiting resistor. However, we are using a variable power supply, so there is no need to protect R3 and thus our actual physical circuit does not include R1.



IMG 1. *optocoupler testing circuit used in this lab*



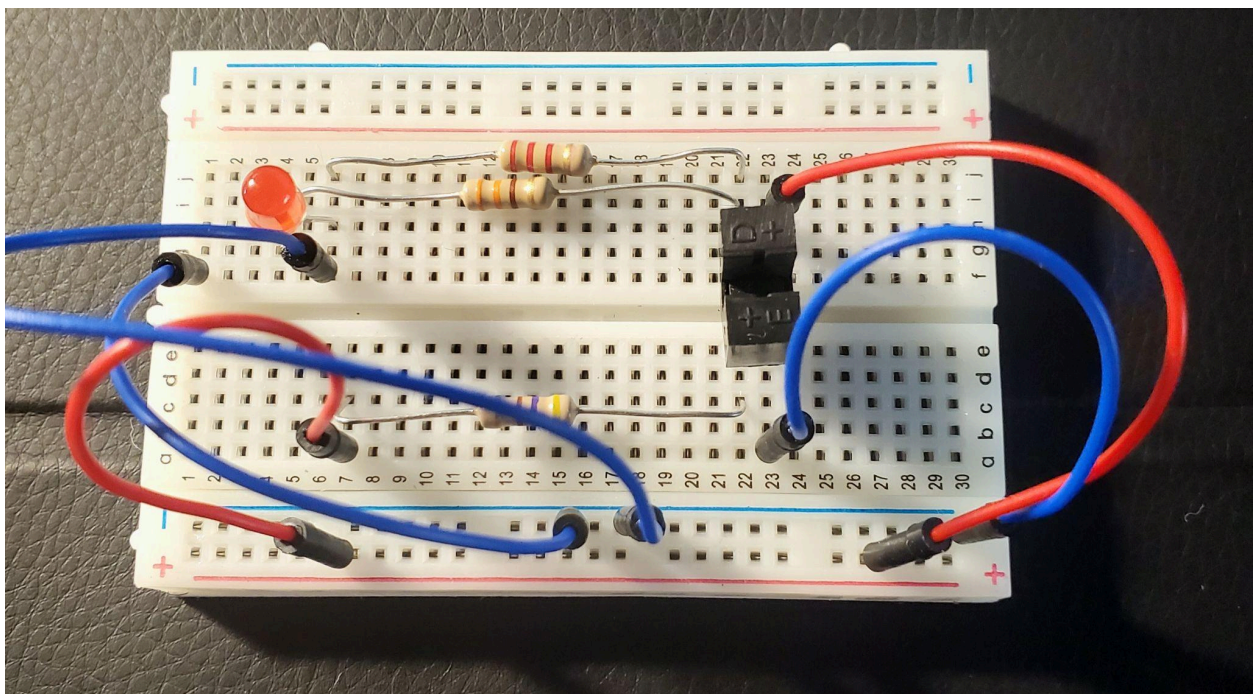
Octocouple r	R1 value	V1 volta ge	V2 voltage
6-Pin DIP Schmitt Trigger Output Optocouple r	280	9.87 v	9.88v

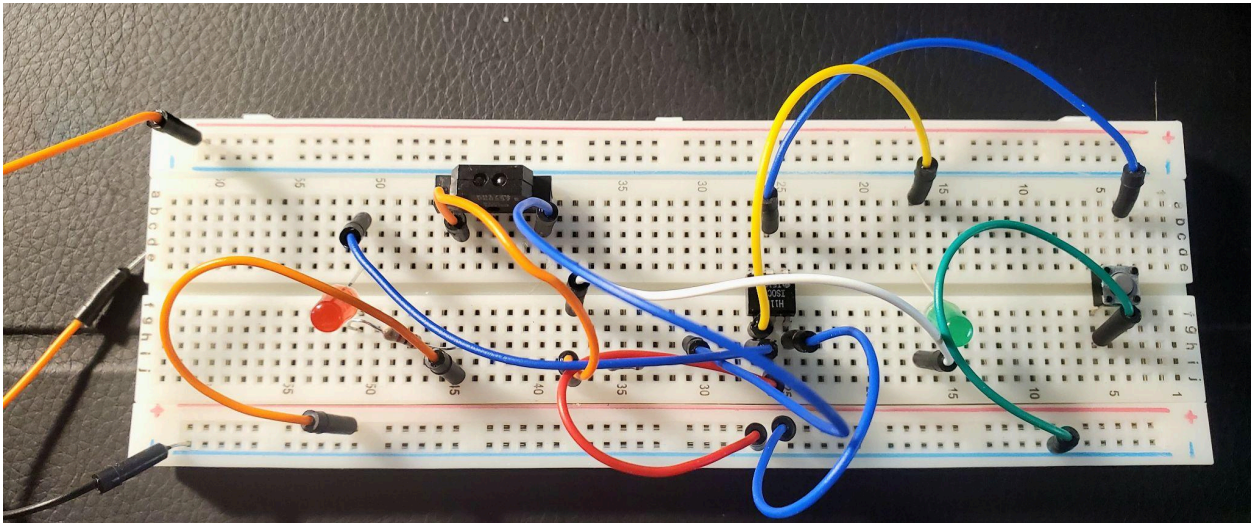
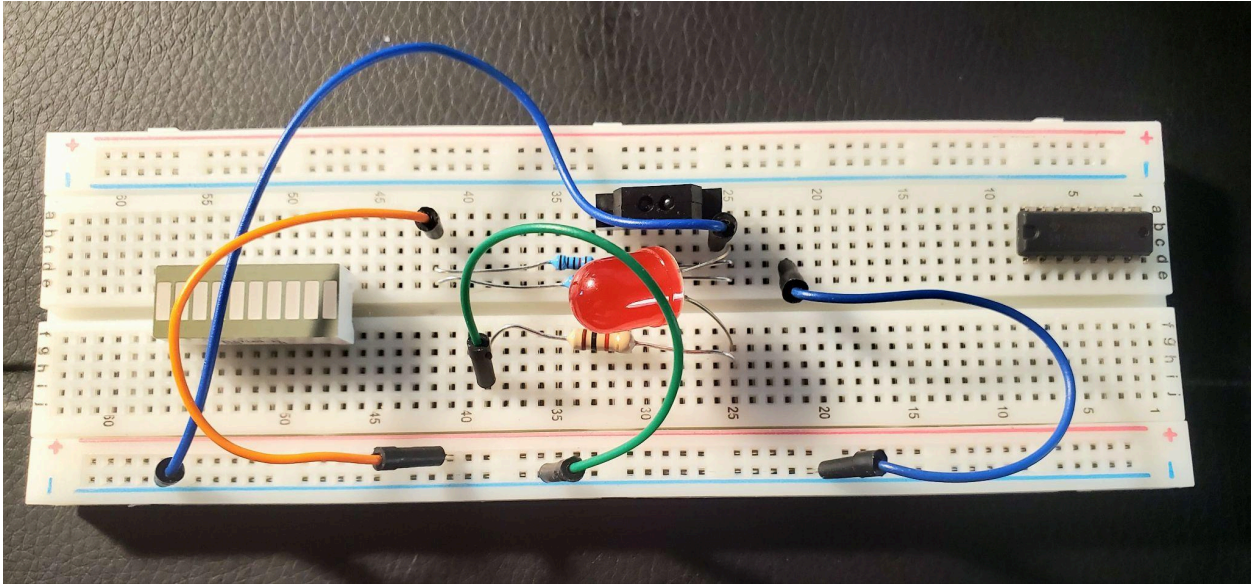


Input: 10 V

Amps: 0.182

1.8 watts





Conclusion

In this lab, we focused on optocouplers and their role in providing electrical isolation between different parts of a circuit. Optocouplers, use light to transmit signals between components while keeping them electrically separated. This is especially important in systems where high voltages or noisy signals could damage sensitive components or create safety risks. A common example of an optocoupler is the 4N36, which consists of an LED and a phototransistor inside a single package.

We worked with several types of optocouplers during this lab:

- A) IC-style optoisolators like the 4N36, ICPL1631, and H11L1
- B) Slotted opto-interrupters such as the H22A1, OPB970, OPB854A1, and OPB811L55
- C) Reflective optocouplers similar to the Honeywell HOA0149, which were custom wired for our setup

Unfortunately, both my lab partner and I were sick during the scheduled lab session, so we had to complete the circuits and testing from home. Despite the challenge, we successfully built and tested the circuits using the optocouplers listed above. This experience reinforced our understanding of how optocouplers isolate control and power sections of a system, making them essential for protecting microcontrollers, sensors, and other low-voltage components from high-voltage circuits.